Assignment on Verilog Programs

1. Write a VERILOG program using BEHAVIOURAL model to implement the following:
    a. 3:8 decoder
    b. BCD to 7 segment display
    c. 4:1 muliplexer
    d. 8:1 muliplexer
    e. 1:4 demux
    f. 1:8 demux
2. Write a VERILOG program using STRUCTURAL model to implement the following:
    a. Implement the SOP function Y = ABC+AB+A
    b. Implement the POS function Y = (A+B+D)(A+C)(A+B)
    c. BCD to XS3 converter
    d. Half adder
    e. Half subtractor
    f. Full adder (do not use module used in 2c)
    g. Full subtractor (do not use module used in 2e)
    h. 4-bit binary carry propagate adder (use full adder module used before in 2e)
    i. 4-bit Binary adder cum subtractor
    j. 3:8 decoder
    k. 4:16 using 3:8 decoder
    l. 5: 32 using four 3:8 and one 2:4 decoders
    m. Implement a full adder using 4:8 decoder
    n. Implement $F(a,b,c) = \sum m(4,6,7)$ using 1:2 decoder
    o. Implement $F(a,b,c) = \sum m(0,1,2,3,6)$ using 2:4 decoder
    p. Design a octal to binary encoder
    q. Design a gray to binary encoder
    r. Design a 4:1 mux using 2:4 decoder
    s. Implement 8:1 mux from 4:1 mux
    t. Implement 4:1 mux from 2:1 mux
    u. Implement $F(a,b,c) = \sum m(1,3,4,11,12,13,14,15)$ using 8:1 mux
    v.
3. Write a VERILOG program using DATA FLOW model to implement the following:
    a. 9's complement of a number: input & output must be decimal
    b. 10's complement of a number: : input & output must be decimal
    c. BCD subtraction using 9's complement: input & output must be decimal
    d. BCD subtraction using 10's complement: input & output must be decimal
    e. XS3 subtraction using 9's complement: input & output must be decimal
    f. XS3 subtraction using 10's complement: input & output must be decimal
    g. 1's complement of a number: input & output must be binary
    h. 2's complement of a number: input & output must be binary
    i. Perform binary addition
    j. Perform binary multiplication

k. Perform binary division by 2
l. Perform BINARY to GRAY conversion
m. Perform GRAY to BINARY conversion
n. Perform 12 bit even parity Hamming code
o. Perform 15 bit even parity Hamming code
p. Implement the function Y = ABC+AB+A
q. Implement the POS function Y = (A+B+D)(A+C)(A+B)
r. Half adder
s. Half subtractor
t. Full adder
u. Full subtractor
v. BCD to XS3 converter
w. XS3 to BCD comverter
x. 1-bit comparator
y. N bit comparator, where N is the last digit of your roll number, if N=0 to 1, add 5 to N.
z. 3:8 decoder
aa. 4:1 muliplexer